Pandas Notes - Part 1

This PDF covers beginner Pandas functions with explanation and examples.

1. read_csv()

Purpose: Reads a CSV file into a DataFrame.
Syntax: df = pd.read_csv('file.csv')
Use when your data is stored in CSV format.

2. read_excel()

Purpose: Reads Excel files.
Syntax: df = pd.read_excel('file.xlsx')

3. head()

Purpose: Displays the first 5 rows.
Syntax: df.head() or df.head(10)

4. tail()

Purpose: Displays the last rows.
Syntax: df.tail()

5. shape

Purpose: Returns (rows, columns).
Example: df.shape

6. columns

Purpose: Displays all column names.
Example: df.columns

7. info()

Purpose: Shows data types, non-null counts and memory usage.
Example: df.info()

8. describe()

Purpose: Summary statistics for numeric columns.
Example: df.describe()

9. sample()

Purpose: Returns random rows.
Example: df.sample(5)

10. loc[]

Purpose: Label-based indexing.
Example: df.loc[0:5]

11. iloc[]

Purpose: Position-based indexing.
Example: df.iloc[0:5,0:3]

12. Filtering

Example: df[df['amount']>100]
Multiple conditions: df[(df['amount']>100) & (df['country']=='US')]

13. isnull()

Purpose: Find missing values.
Example: df.isnull().sum()

14. dropna()

Purpose: Remove rows with missing values.

15. fillna()

Purpose: Fill missing values.
Example: df.fillna(0)

16. value_counts()

Purpose: Count occurrences.
Example: df['country'].value_counts()

17. unique() and nunique()

Purpose: Show unique values and count unique values.

18. groupby()

Purpose: Group data for aggregation.
Example: df.groupby('country')['amount'].mean()

19. sort_values()

Purpose: Sort data.
Example: df.sort_values('amount', ascending=False)

20. corr()

Purpose: Correlation between numeric columns.
Example: df.corr(numeric_only=True)

Practice Tips

1. Load a dataset.
2. Use head(), info(), describe().
3. Check missing values.
4. Filter data.
5. Use groupby() and value_counts().
6. Build EDA before machine learning.